

Data Security

Master's in Cybersecurity

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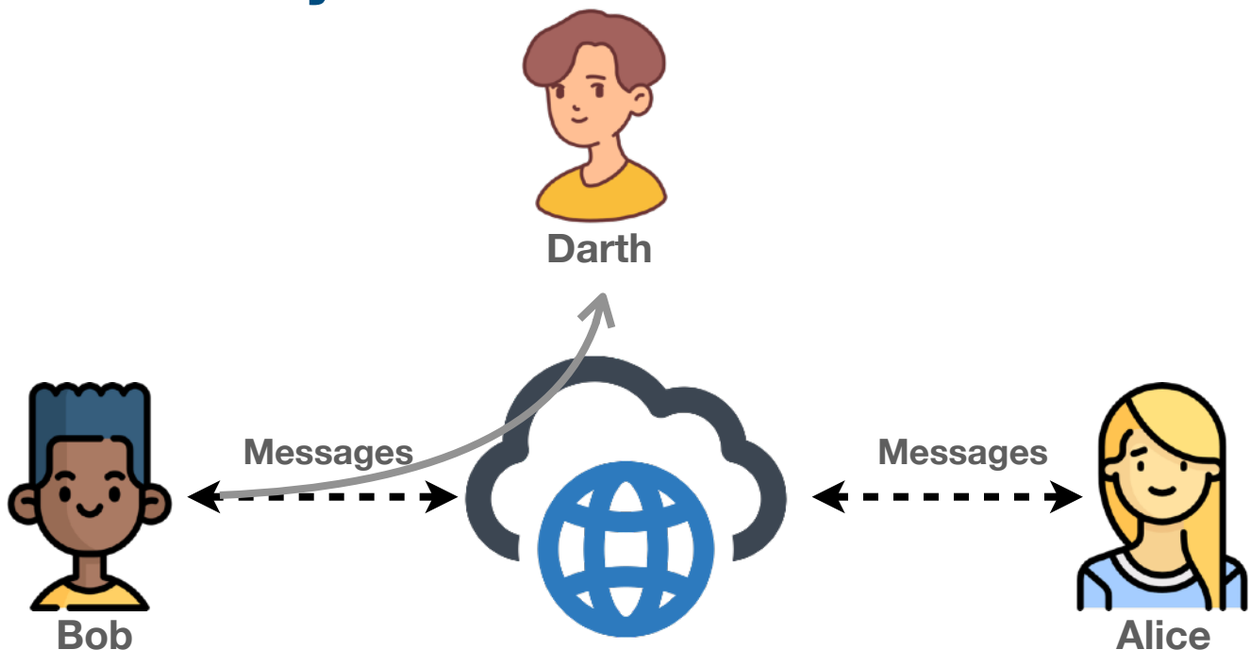
Data Security

Objectives

- Understand the basic concepts of modern cryptography
- Understand the fundamentals of symmetric cryptography
- Comprehend the practical concepts regarding the security of cryptographic schemes
- Understand classic cryptography techniques and their vulnerabilities to different cryptanalysis strategies

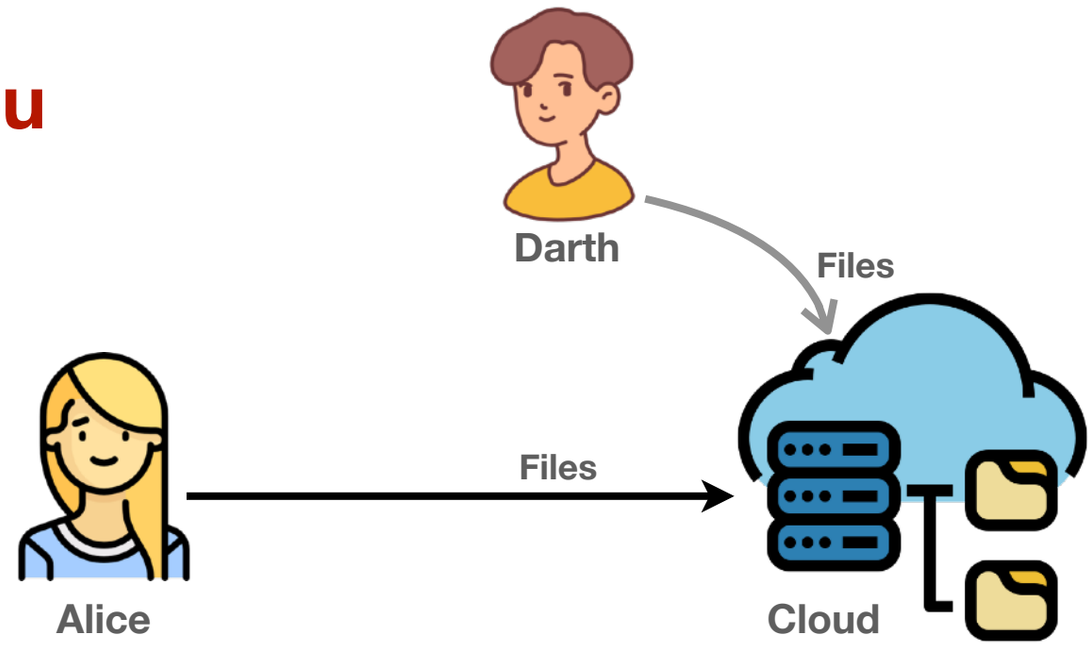
Principles

How to provide security in both scenarios?



What exactly are you trying to protect?

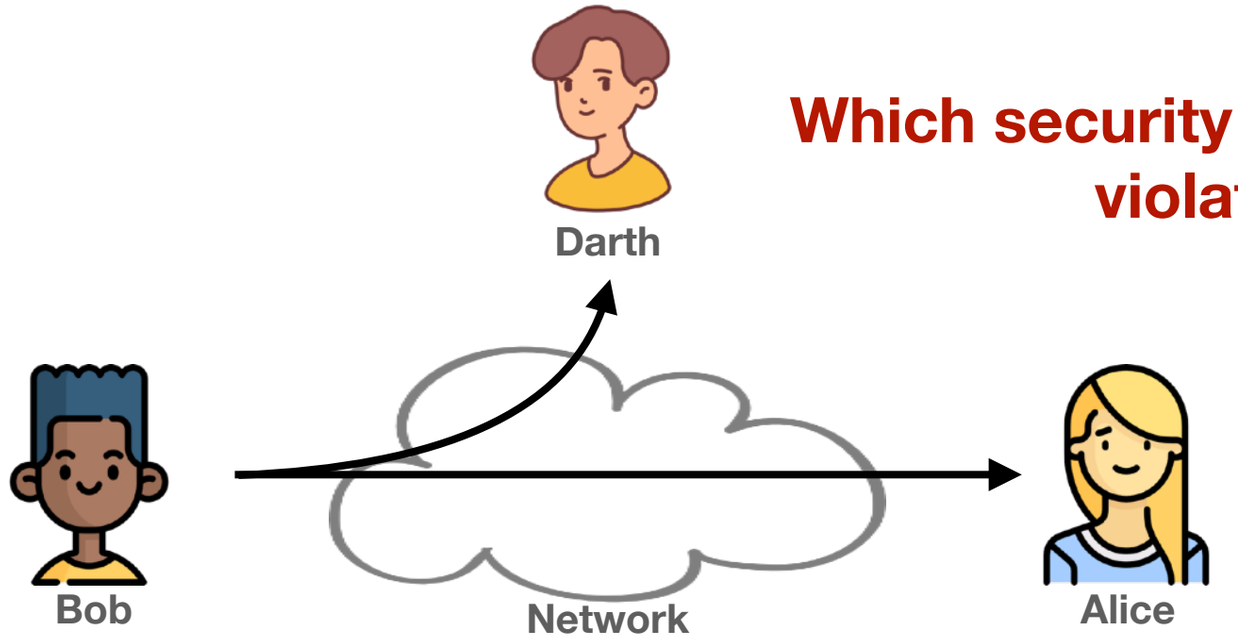




Principles

Attacks

Passive attacks: eavesdropping or monitoring transmissions without any alteration of the data.



Which security properties are violated?

- **Release of message contents**
- **Traffic analysis**

Passive attacks are challenging to detect, as there is no change to the messages' content. Their prevention usually involves cryptographic techniques.

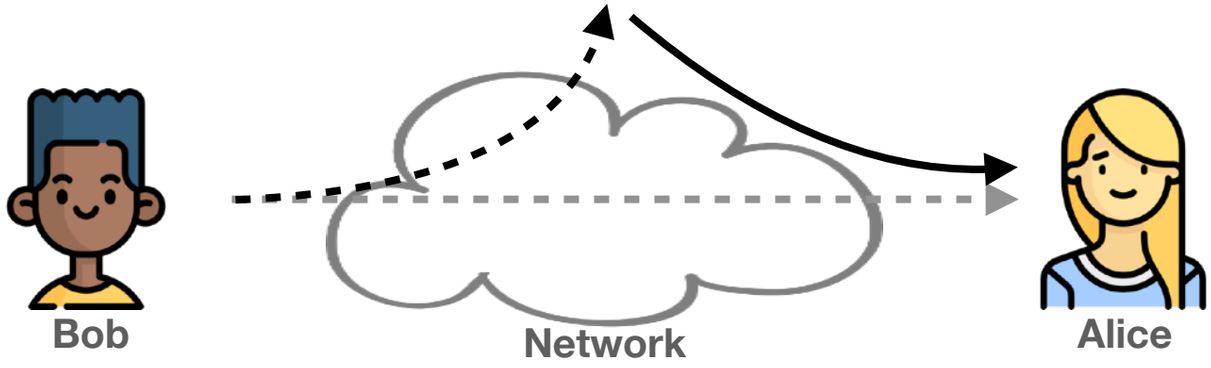
Principles

Attacks

Active attacks: involve some modification of the data stream or the creation of a false stream.


Darth

Which security properties are violated?



- **Masquerade**
- **Replay**
- **Modification of messages**
- **Message destruction**

Usually, in addition to prevention (which is difficult due to the large number of attack types), the primary defense strategy is to detect active attacks.

Data Security

Applied cryptography

What is Cryptography?

The discipline that embodies the principles, means, and methods for the transformation of data to hide their semantic content, prevent their unauthorised use, or prevent undetected modification.

NIST SP 1800-21B

It aims at protecting security properties

- Confidentiality
- Integrity
- Availability
- Authenticity
- Non-repudiation

Cryptography

Principles

Plaintext: the original message or data fed into a cryptographic algorithm as input.

Encryption algorithm: a set of mathematically expressed rules for rendering data unintelligible by executing a series of conversions controlled by a key.

CNSSI 4009-2015

Secret key: an input the encryption algorithm uses to perform the substitutions and transformation in the plaintext.

Ciphertext: the scrambled message produced as output. It depends on the plaintext and the secret key. For a given message, two different keys will produce two different ciphertexts.

Decryption algorithm: a set of rules that take the ciphertext and the secret key to produce the original plaintext. This is essentially the encryption algorithm run in reverse.

In symmetric cryptography, the encryption and decryption algorithms use the same key.

Cryptography

Principles

Cryptanalysis: the study of mathematical techniques for defeating cryptographic techniques without an initial knowledge of the used key. This includes looking for errors or weaknesses in the implementation of an algorithm or the algorithm itself.

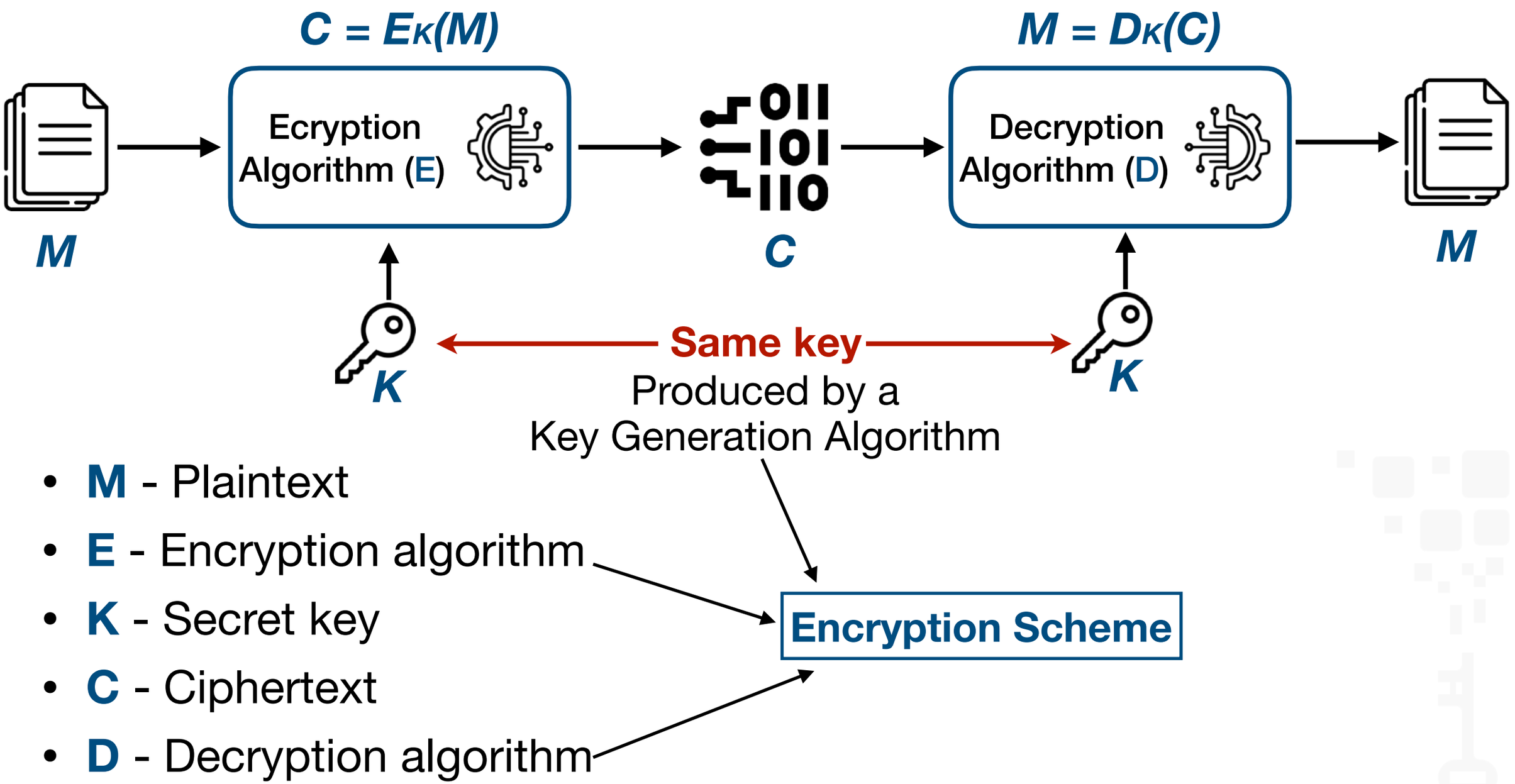
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Cryptology: the science that deal with cryptography and cryptanalysis.

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Cryptography

Simplified Model of Symmetric Encryption



- **M** - Plaintext
- **E** - Encryption algorithm
- **K** - Secret key
- **C** - Ciphertext
- **D** - Decryption algorithm

Cryptography

Symmetric Encryption operation

Block Ciphers: the cryptographic algorithm operates on fixed-length blocks of the input (*e.g.,* plaintext) with the resulting output block (*e.g.,* ciphertext) of the same size.

- Common block sizes are 64 and 128 bits

Stream Ciphers: the cryptographic algorithm processes an individual bit, byte, or character of the input (*e.g.,* plaintext) at a time.

Substitution: each plaintext element or group of elements is uniquely replaced by a corresponding ciphertext element or group of elements

Permutation: a sequence of plaintext elements is replaced by a permutation of the sequence without adding or removing any element from the original sequence.

Cryptography

Principles

Defining the security of cryptographic schemes

Regardless of any prior information an attacker has about the plaintext, the ciphertext should reveal no additional information about the plaintext.

An encryption scheme is **unconditionally secure (*perfect secrecy*)** if the ciphertext generated by the scheme does not contain enough information to recover the corresponding plaintext, no matter how much ciphertext is available and how much time an attacker has.

An encryption scheme is **computationally secure (*computational secrecy*)** if the ciphertext generated by the scheme meets one or both of the following criteria:

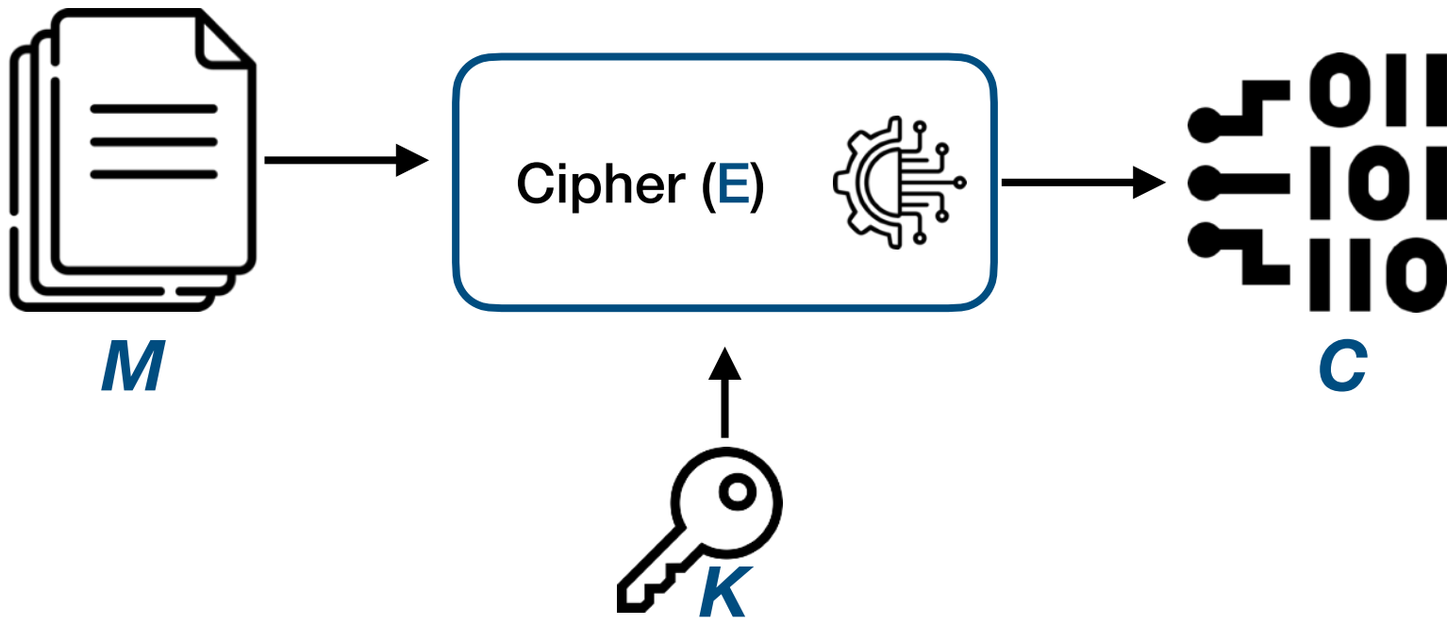
- The cost of breaking the cipher exceeds the value of the encrypted information;
- The time required to break the cipher exceeds the useful lifetime of the information.

Cryptography

Kerckhoff's Principle

A cryptosystem should be secure even if everything about the system, except the key, is public knowledge.

$$C = E_K(M)$$



Its security is ensured by the Secret Key

Cryptography

Attacks on Cryptographic Systems

The objective is either to recover the plaintext or the secret key in use.

Cryptanalysis

Rely on the nature of the algorithm plus perhaps some knowledge of the general characteristics of the plaintext or even some sample plaintext-ciphertext pairs. The known characteristics are used to deduce a specific plaintext or the secret key.

Brute-force attack / Exhaustive search attack

The attacker tries every possible key on a piece of ciphertext until an intelligible translation into plaintext is obtained. On average, half of all possible keys must be tried to achieve success.



Many cryptography-related vulnerabilities are not caused by the weakness of the encryption algorithms, but by the incorrect use or implementation of those algorithms.

Cryptography

Attacks on Cryptographic Systems

In most network situations, we can assume that the ciphers are known. What generally makes brute-force cryptanalysis impractical is the use of an algorithm that employs a large **key space**.

Estimated time for exhaustive search

Key size	**Time** (1*μsec/test*)	**Time** (1*μsec/*10^6*test*)
32 bit	35.8 min	2.15 msec
40 bit	6.4 days	550 msec
56 bit	1140 years	10 hours
64 bit	500000 years	107 days
128 bit	$5x10^{24}$ years	$5x10^{18}$ years

- Note that increasing the key size by one bit doubles the available key space.
- Currently, 2^{80} is considered to provide an acceptable level of security.

Cryptography

Cryptanalysis strategies

Type of Attack	Known to Cryptanalyst
Ciphertext Only	■ Encryption algorithm ■ Ciphertext
Known Plaintext	■ Encryption algorithm ■ Ciphertext ■ One or more plaintext–ciphertext pairs formed with the secret key
Chosen Plaintext	■ Encryption algorithm ■ Ciphertext ■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	■ Encryption algorithm ■ Ciphertext ■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	■ Encryption algorithm ■ Ciphertext ■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key ■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Classical Techniques

Caesar Cipher

A **substitution** cipher allegedly created by Julius Caesar during the Gallic campaign.

It works by replacing each letter of the alphabet with the letter standing three places further down the alphabet.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Example

Plaintext: attackatnineam

Ciphertext: DWWDFNDWQLQHDP

26 possible keys.
One of them is insecure.

It is possible to use an arbitrary offset (key).

Plaintext: attackatnineam

Ciphertext (k=6): GZZGIQGZTOTKGS

Classical Techniques

Caesar Cipher

Algorithm

By assigning a numerical equivalent to each letter, the algorithm can be expressed as follows:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

For each plaintext *p*, substitute the ciphertext letter *C*:

$$C = E(k,p) = (p + k) \bmod 26$$

The decryption algorithm is:

$$p = D(k,C) = (C - k) \bmod 26$$

Classical Techniques

Caesar Cipher

Cryptanalysis

- The encryption and decryption algorithms are known
- There are only 26 keys to try
- The language of the plaintext is known and easily recognisable

Ciphertext: XQQXZHXQKFKBXJ

Key = 1: YRRYAIYRLGLCYK

Key = 2: ZSSZBJZSMHMDZL

Key = 3: ATTACKATNINEAM

Key = 4: BUUBDLBUOJOFBN

...

Key = 24: VOOVXFVOIDIZVH

Key = 25: WPPWYGWPJEJAWI

Easy brute-force cryptanalysis

Classical Techniques

Caesar Cipher

Cryptanalysis

- The encryption and decryption algorithms are known
- There are only 26 keys to try
- The language of the plaintext is known and easily recognisable

The plaintext output may not be recognizable if the plaintext language is unknown.

Example

Original plaintext: attackatnoonafterthesecondringofthechurchbells

Compressed plaintext: ILIOHNMKKOr3J3RsrPfz007Wfz00

Classical Techniques

Monoalphabetic Ciphers

- Enhances Caesar cipher by allowing arbitrary substitutions (i.e., *permutations*)
- A single cipher alphabet (mapping from plain alphabet to cipher alphabet) is used per message
- Increases the key space

Example

Consider the finite set of elements $S = \{a,b,c\}$

There are six permutations of S: *abc, acb, bac, bca, cab, cba*

What are the number of possible permutations for the alphabet?

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

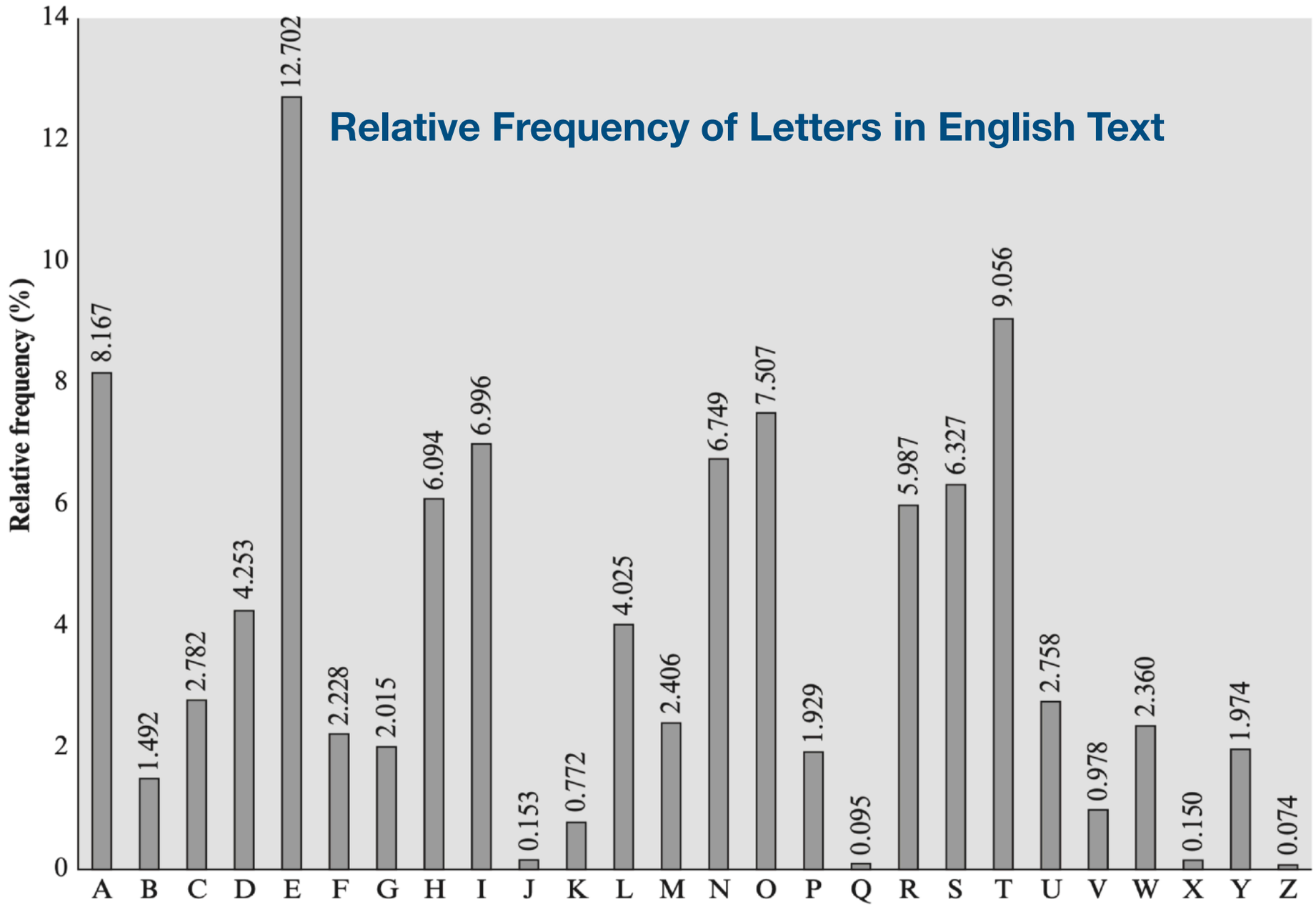
In general, there are $n!$ Permutations of a set of n elements

$26! > 4 \times 10^{26}$ → Strong against brute-force cryptanalysis

Classical Techniques

Monoalphabetic Ciphers

Cryptanalysis



Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Classical Techniques

Monoalphabetic Ciphers

Cryptanalysis - Example

Ciphertext

```
UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ
```

Bigrams and trigrams are also useful

The relative frequencies of the letters in the ciphertext

P 13.33	H 5.83	F 3.33	B 1.67	C 0.00
Z 11.67	D 5.00	W 3.33	G 1.67	K 0.00
S 8.33	E 5.00	Q 2.50	Y 1.67	L 0.00
U 8.33	V 4.17	T 2.50	I 0.83	N 0.00
O 7.50	X 4.17	A 1.67	J 0.83	R 0.00
M 6.67				

Use this to recover the plaintext!

Source: Stallings, W. & Brown, L. Cryptography and Network Security: Principles and Practice

Classical Techniques

Monoalphabetic Ciphers

Cryptanalysis - Result

Ciphertext

```
UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ
```

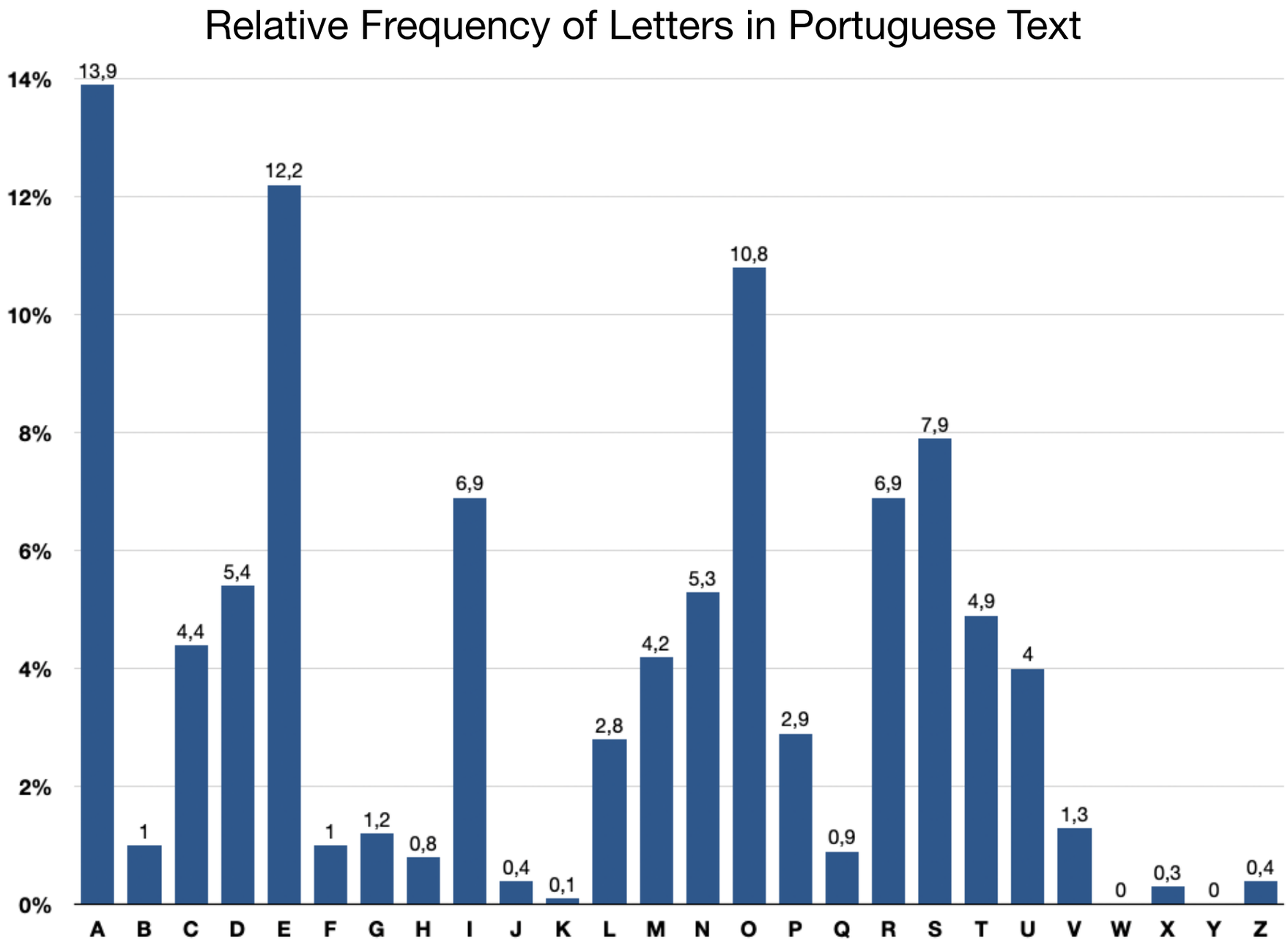
Plaintext

```
it was disclosed yesterday that several informal but
direct contacts have been made with political
representatives of the viet cong in moscow
```

Classical Techniques

Monoalphabetic Ciphers

Cryptanalysis



Classical Techniques

Polyalphabetic Ciphers

Vigenère cipher

- Introduced by Blaise Vigenère (late 16th century) and known as “le *chaffer indéchiffrable*”.
- It consists of 26 Caesar Ciphers with shifts of through 25.
- It was broken at the end of the 19th century by Charles Babbage and Friedrich Kasiski.

Uses a different substitution a different positions in the message.

Description

- The Key is a "phrase" in which each letter determines a substitution.
- Each substitution is a simple shift determined by the respective character in the key (*e.g.*, `A=0, B=1, …, Z=25`).
- The Key size defines the number of substitutions that are used.

Classical Techniques

Polyalphabetic Ciphers

Vigenère cipher - example

Original plaintext: attackatnoonafterthesecondringofthechurchbells

Key: SECRET

Ciphertext: SXVRGDSXPFSGSJVVVMZIUVGHFHTZRZGJVYIVZYTTLUWPNJ

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Original plaintext:	a	t	t	a	c	k	a	t	n	o	o	n
Key:	S	E	C	R	E	T	S	E	C	R	E	T
Ciphertext:	S	X	V	R	G	D	...					

Note that the same input produces different outputs.

Classical Techniques

Polyalphabetic Ciphers

Vigenère cipher - algorithm

By assigning a numerical equivalent to each letter, the algorithm can be expressed as follows:

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

For a plaintext $p = p_0, p_1, p_2, \ldots, p_{(n-1)}$ and a key $k = k_0, k_1, \ldots, k_{(m-1)}$, the general encryption equation for C_i is:

$$C_i = (p_i + k_{i \bmod m}) \bmod 26$$

The decryption generalization is:

$$p_i = (C_i - k_{i \bmod m}) \bmod 26$$

Classical Techniques

Polyalphabetic Ciphers - Vigenère Cipher

Cryptanalysis

Cipher broken at the end of the 19th century by Charles Babbage and Friedrich Kasiski.

- **Infer probable sizes for the key (*e.g.*, by analyzing the spacing between repeated patterns - Kasiski test);**
- **Break the cryptogram into "slices" (one for each letter of the key);**
- **Attack each of them (Caesar cipher) using frequency analysis.**

Explore the Vigenère Cipher's cryptanalysis in detail at this link.

Classical Techniques

Transposition

Transposition ciphers generate ciphertext by performing some sort of permutation on the plaintext letters.

Example - A simple transposition scheme

Original plaintext: **attackatnoon**

Key: **4213** (this is the permutation order)

1	2	3	4
a	t	t	a
c	k	a	t
n	o	o	n

← **Read the matrix following the key order**

Ciphertext: **ATATTKCANONO**

Cryptanalysis

The ciphertext preserves the frequency of the letters from the plaintext.



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